PROJECT- Network Security Cyber Threat Lab Creation

Description:

In today's world, network security is becoming bigger concern day-by-day. As technology is growing and advancing, cyber threat also becoming a major issue. However, cyber security threats are actions committed by malicious individuals with the intention of acquiring data, damaging computer systems, or interfering with operations. It may come from numerous sources, such as hostile nations and terrorist organizations, individual hackers, as well as trustworthy people like contractors or the employees. There are so many cyber threats existing now. They are malware, web-based attacks, web application attacks, phishing, spam, cyber espionage, ransomware etc. For this particular project, I have researched one specific cyber threat which is *Phishing*. Phishing is a type of social engineering scam where the perpetrators trick victims into divulging personal information. It can be through emails, texts, phone calls, website and so on. The purpose of this lab is to be demonstrating and understanding the tools for phishing attacks. The requirement for this lab is:

- VM Fusion Pro – The Virtual machine environment where the VM will be installed.
- Kali Linux VM

The reason behind using Kali Linux for this lab is that Kali Linux have almost all the basic security vulnerability tools and it also support other tools also. For understanding the phishing attacks, I have used the following tools:

1. SOCIAL ENGINEERING TOOLKIT
2. ZPHISHER
3. SOCIALPHISH
4. EVILGINX2
5. GOPHISH
6. SHELLPHISH

For better illustration, I have used my Mozilla Firefox in the virtual machine.

Topology:

For this topology, I have used Microsoft Visio. This application helps to better understand many components of the network and their interconnections. It also illustrates the ways in which every element works together and supports performance management. Below is the network topology for analyzing phishing attack:

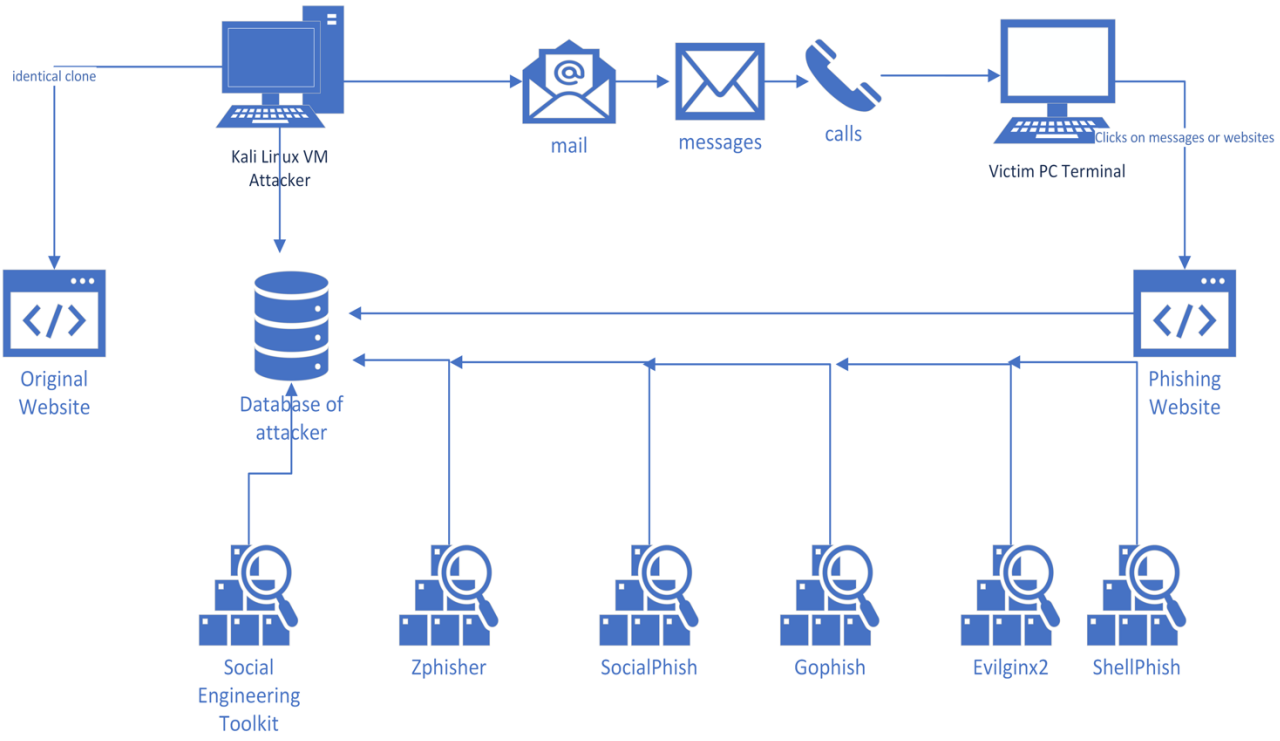


Figure: Network Topology.

Key Syntax:

1) Social Engineering Toolkit (SET): SET is an open-source Python-driven tool aimed at penetration testing around social engineering. In Kali Linux VM by default is has been installed. I have just simply searched the "SET" and get it running by "root".

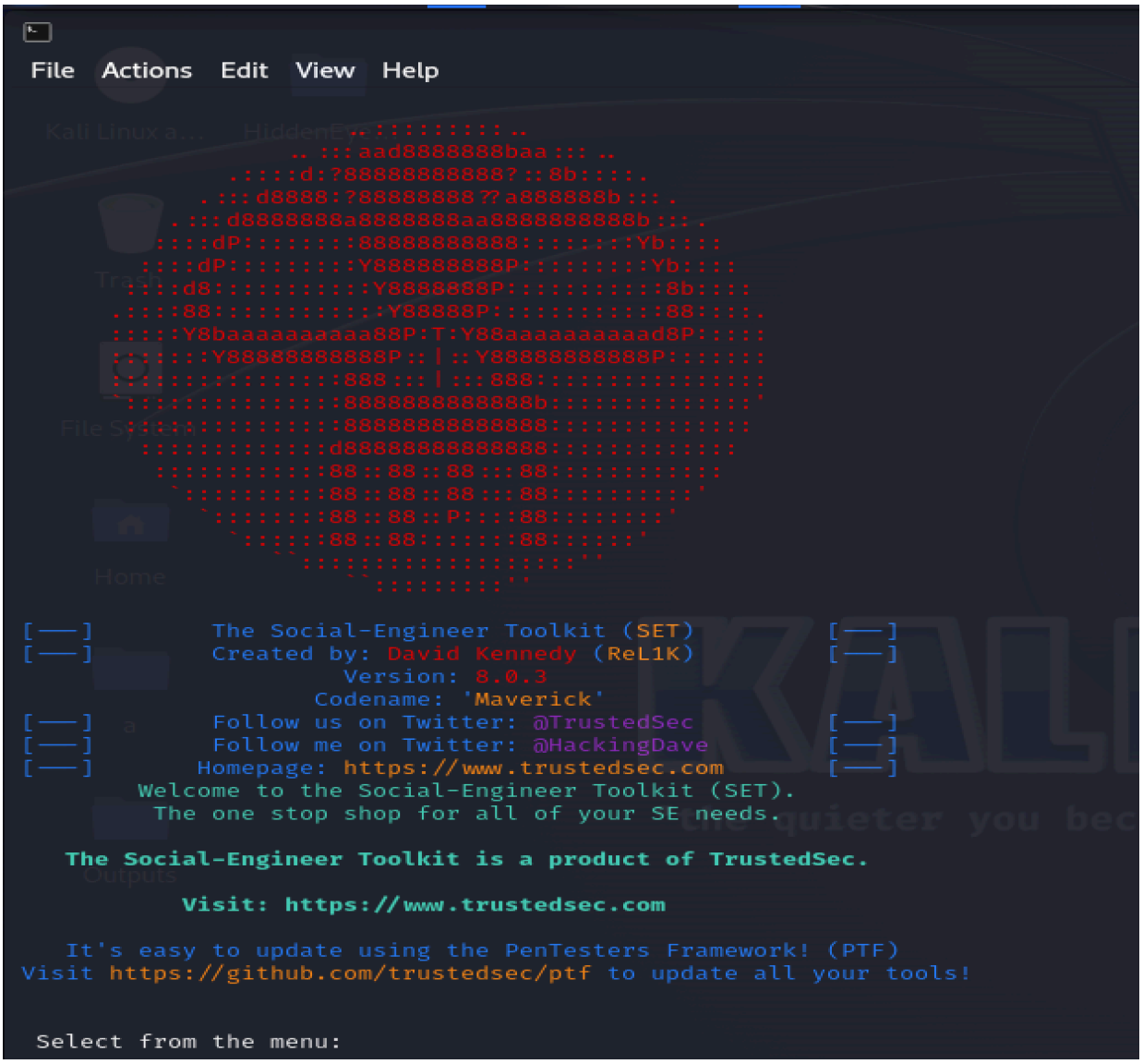


Figure-I: Configuring SET.

2) Socialphish: For configuring Socialphish in Kali Linux, CLI have been used. Here, it is installed by the GitHub clone software, where all modules have been installed shown in Figure-II. The simple command, "git clone https://github.com/pvanfas/socialphish.git" installed the software.



Figure-II: Configuring Socialphish.

3) Gophish: For configuring Gophish in Kali Linux, CLI have been used. Here, it is installed by the GitHub. The simple command, “https://github.com/gophish/gophish.git ” installed the software.



Figure-III: Installing Gophish.

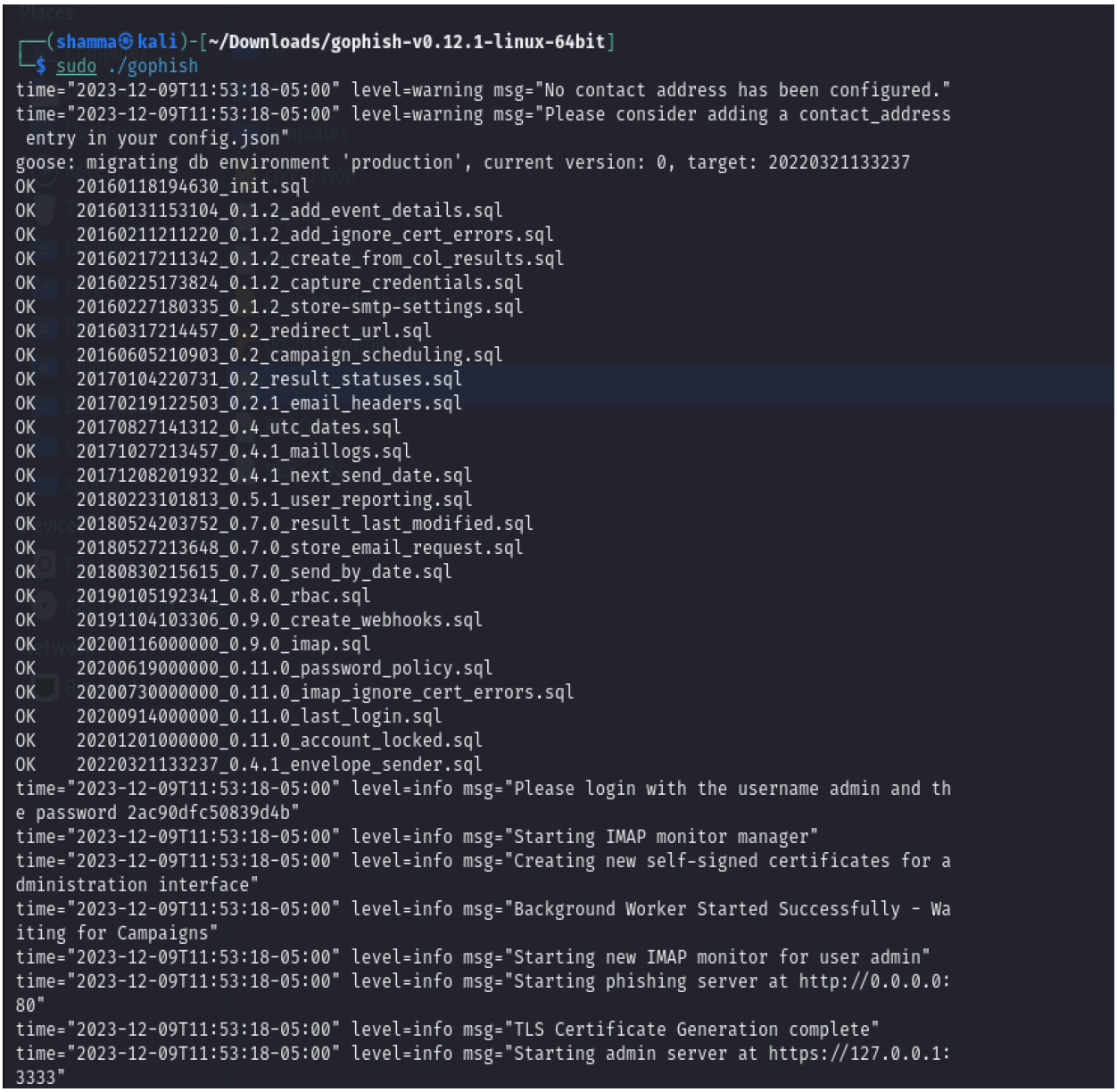


Figure-IV: Configuring Gophish

4) Evilginx2: This tool has been simply installed by “sudo apt install evilginx2”.

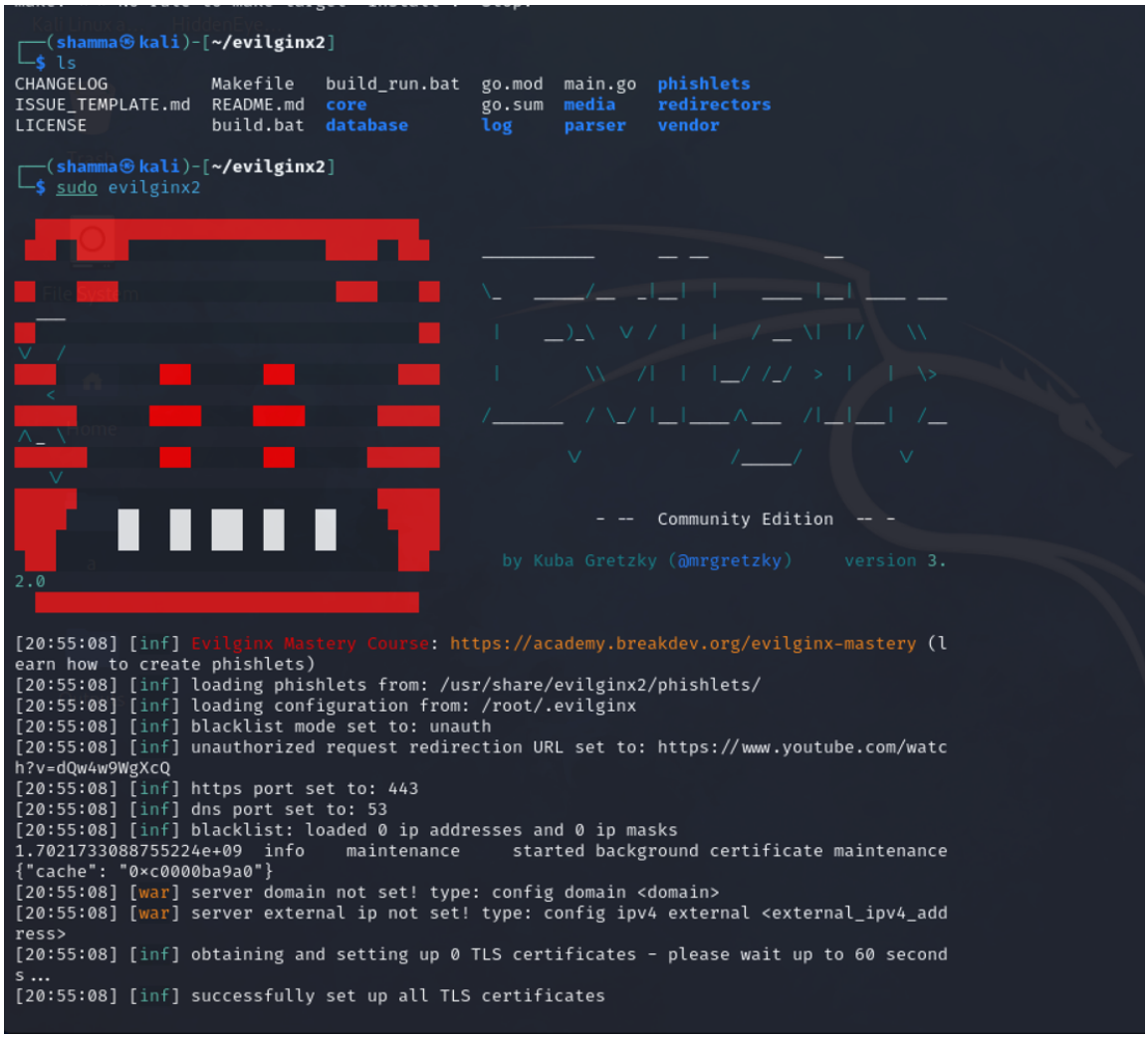
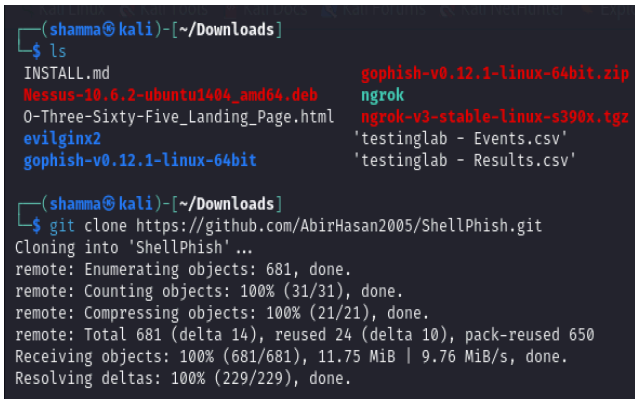


Figure: Configuring Evilginx2

5) Shellphish: This tool has been also installed from GitHub. I have also added the “ngrok” for better results .



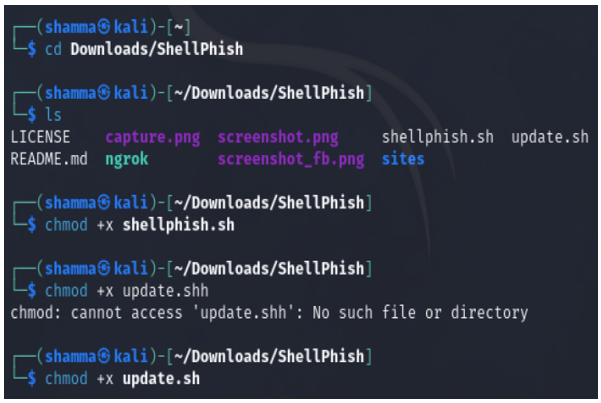


Figure: Installing and configuring Shellphish

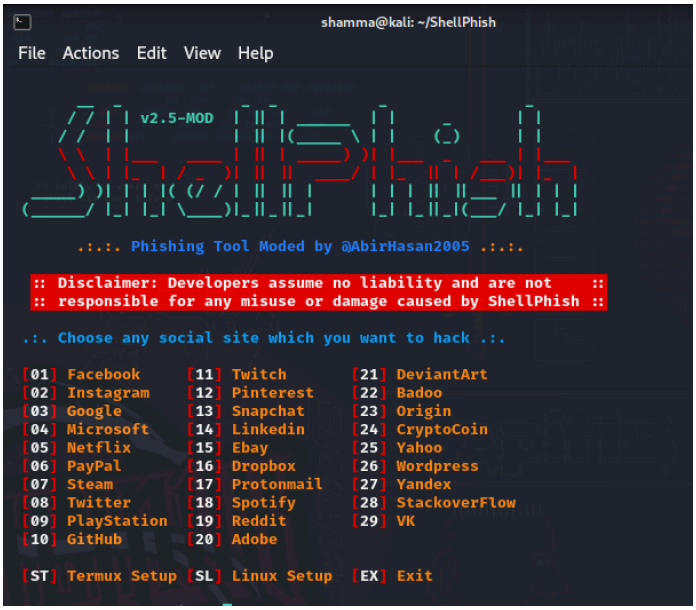


Figure: Shellphish prompt.

Verification:

Task One: Virtual Machine Setup

1) Installed Kali Linux VM in VM Fusion Pro.



Figure: Kali Linux VM.

2) Isolation of the VM: I have isolated the Kali Linux VM by placing the vmnic into my custom internet "vmnet3", whose IP address is 192.168.156.0/24.

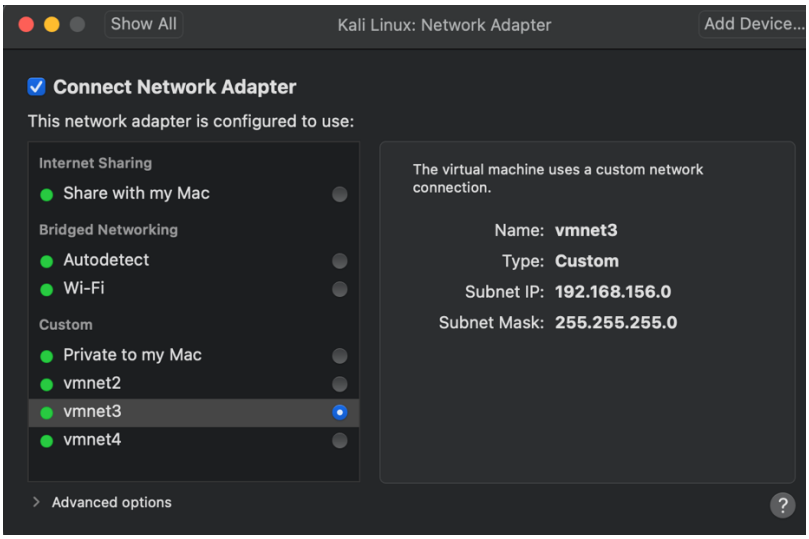


Figure: Network connection of the VM.

TASK TWO: Phishing tools analysis

I have analyzed how phishing attack works by using these tools.

i. **SEToolKit (SET):** With SET's many distinctive attack vectors, constructing a convincing attack is simple and fast. It is imperative for security experts and penetration testers to acquire the correct authorization prior to utilizing these tools in any kind of testing situation. It can be done in many ways. I have illustrated three ways in this lab. They are given below:

   a) **Spear Phishing PDF Creation:** Spear phishing attack is directed at individuals or companies. At SEtoolkit prompt, first select the “social engineering attacks”. Then I have selected option 1, which is spear phishing attack. Next, I have done a mass mail attack, by creating payloads and Metasploit framework which is illustrated in figure-1



Figure-1

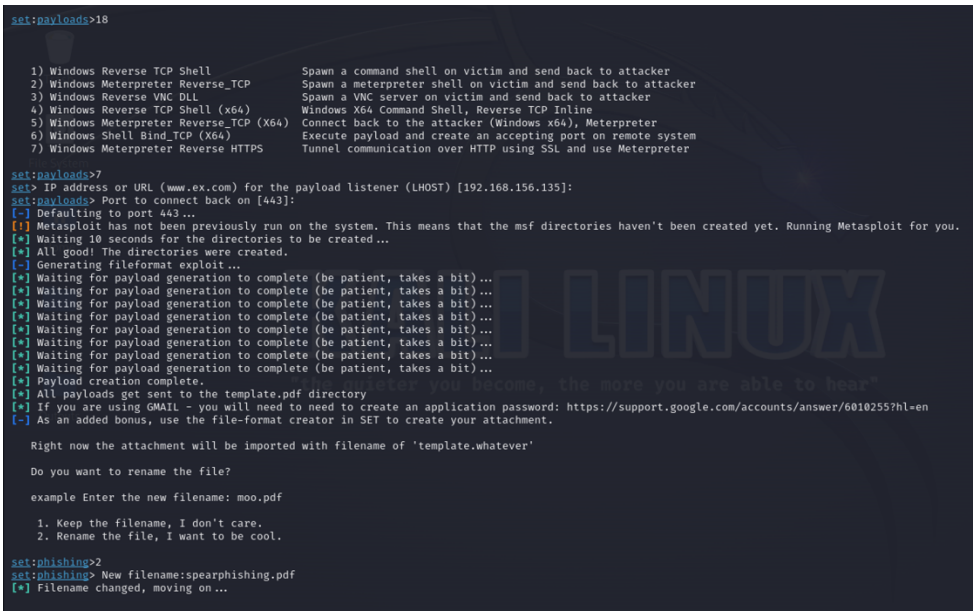
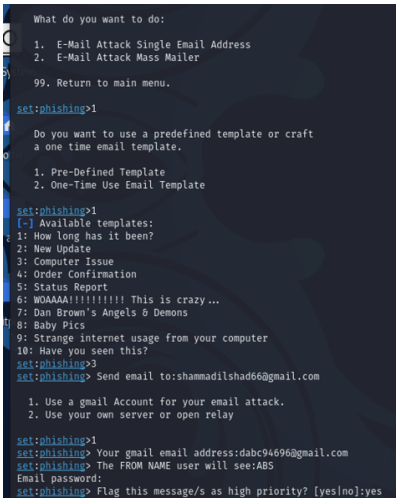


Figure-2

Later on, I have selected the “windows meterpreter reverse https” which allowed me have port listener and then all the directories were created. I have created a pdf file named “spearphishing.pdf”, shown in figure-2.

In figure-3, I have given the email addresses for the phishing attack. Then I have setup a listener.



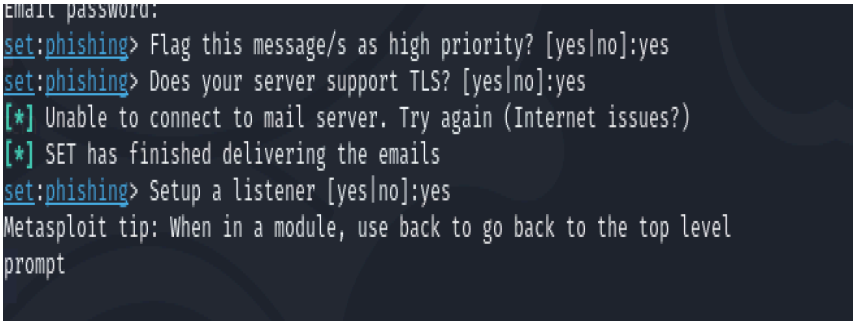


Figure-3 Figure-4

After that, the url provided shows this output below, which simply demonstrates that the task is generated successfully.

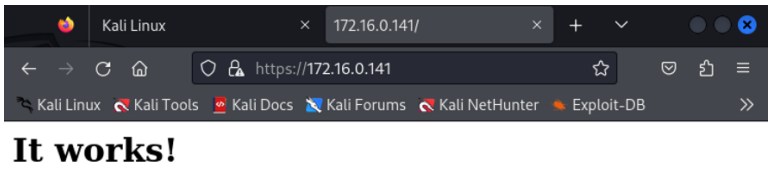


Figure-5

This is the outcome of creating the spearphishing pdf. However, this pdf can also be used to attack in the mail individually (as shown in figure-6).

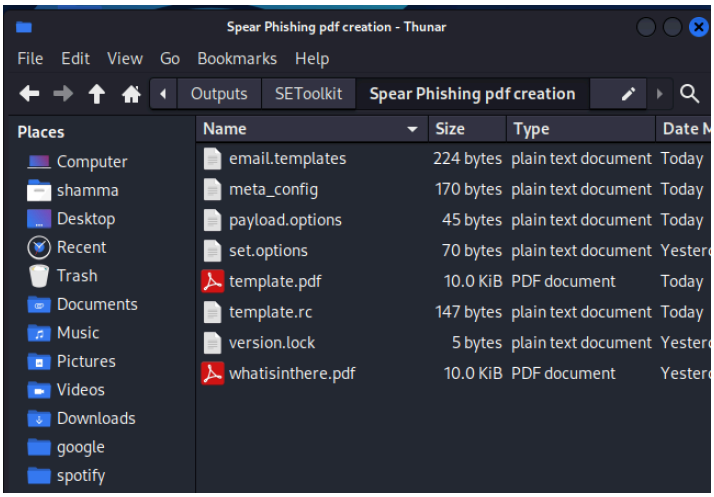


Figure-6

b) **Website Attack Vendors:** Website attack vectors defines the way of attacking a website. It can be in many ways. Below I have illustrated two of them.

- *Site Clone:* Site cloning basically means the act of building a duplicate of an already-existing website, or more simply said, a copy of a website or its associated elements and design, is known as website cloning. In the very first step, I have selected the website attack vendors and then selected the credential harvester attack method which will give me the victim IP address and password (figure-7 and figure --

-8).

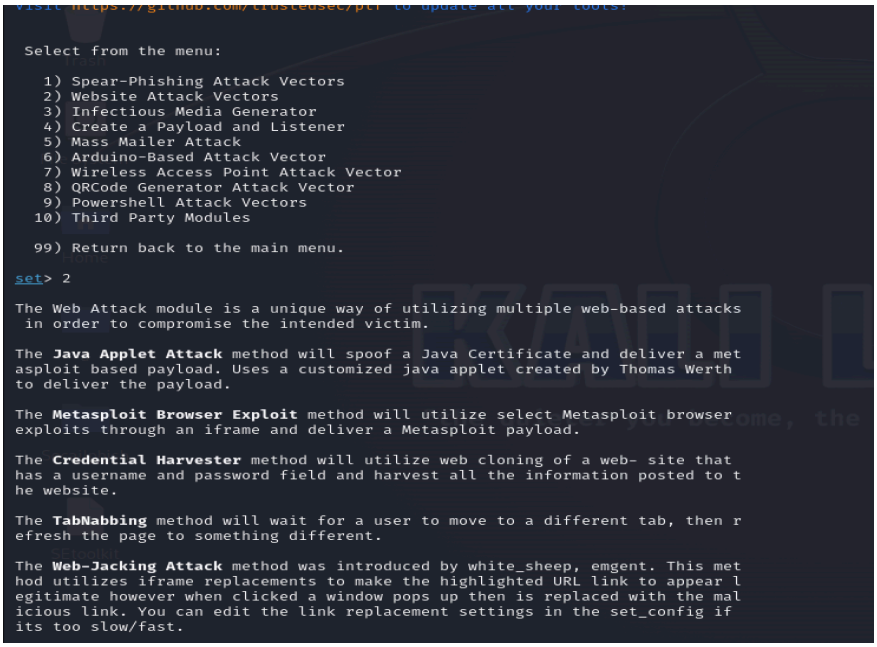


Figure-7

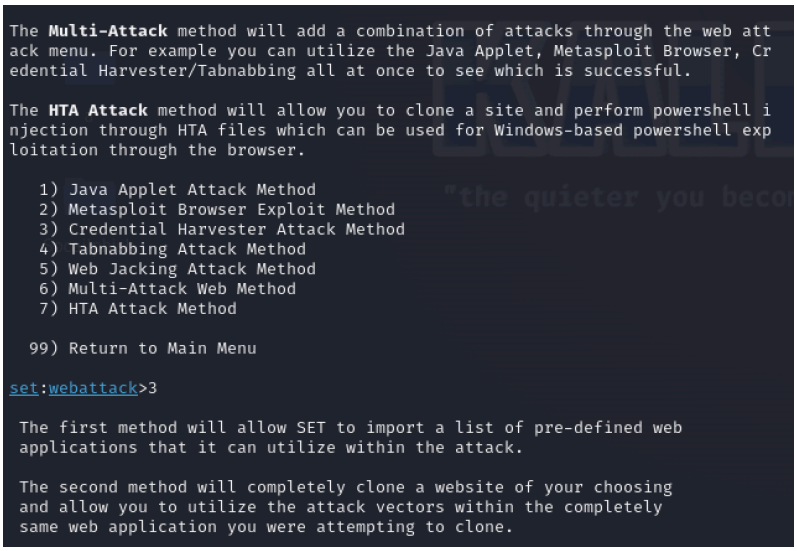


Figure-8

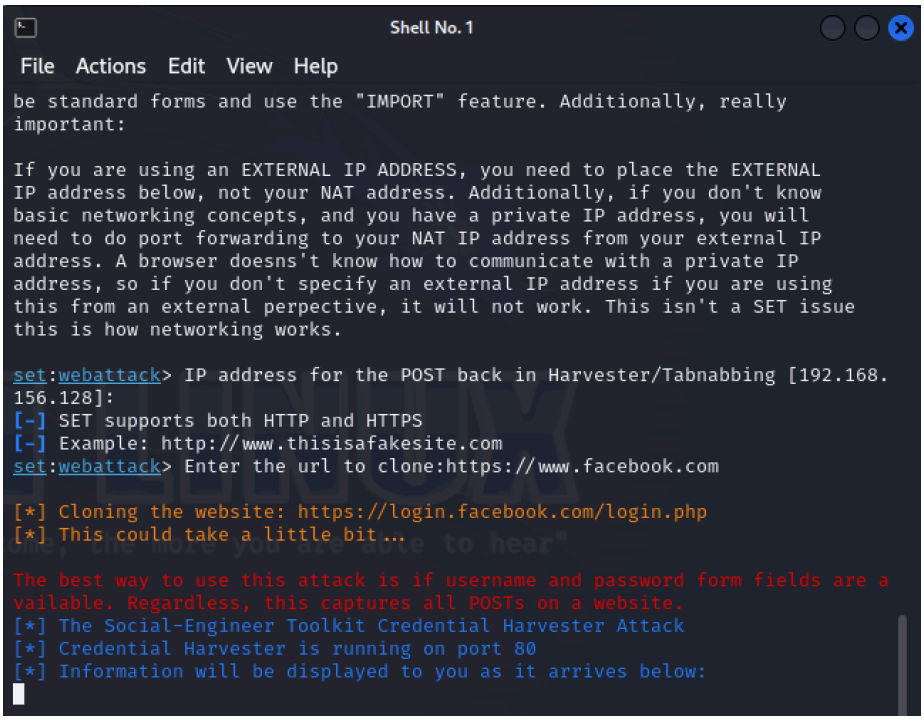


Figure-9

Then, I have cloned the Facebook site by entering the original URL. Then attack vectors clones the website as shown in figure-10. For that, I just needed to copy the generated IP address in my local machine.

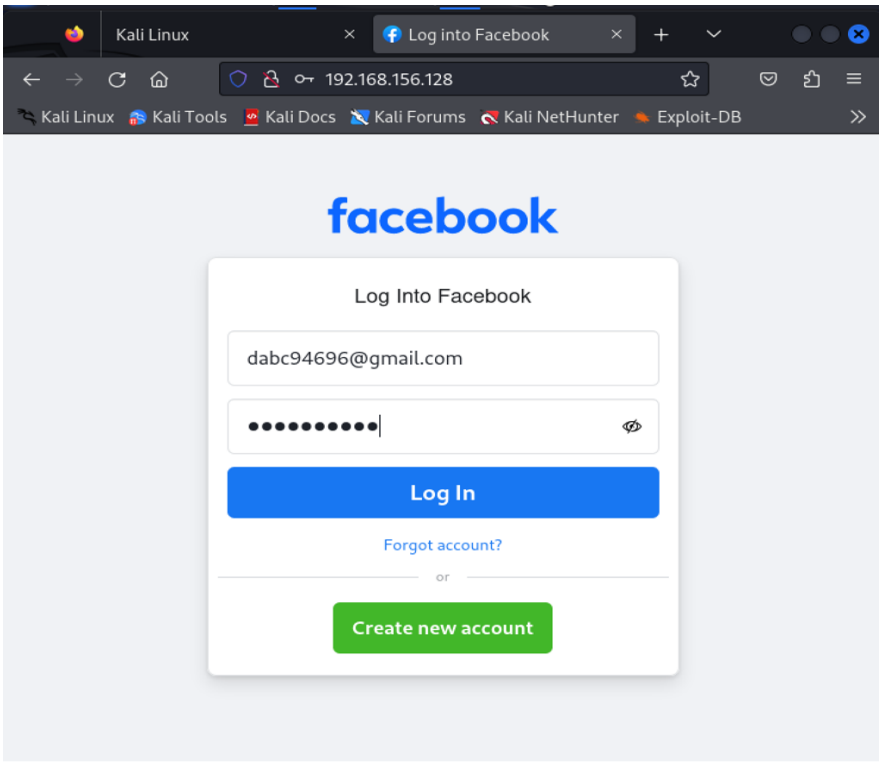


Figure-10

- *Web Templates*: Another way of website attack vendors is creating website templates. For this, I again selected the credential harvester attack method as shown in figure-10.

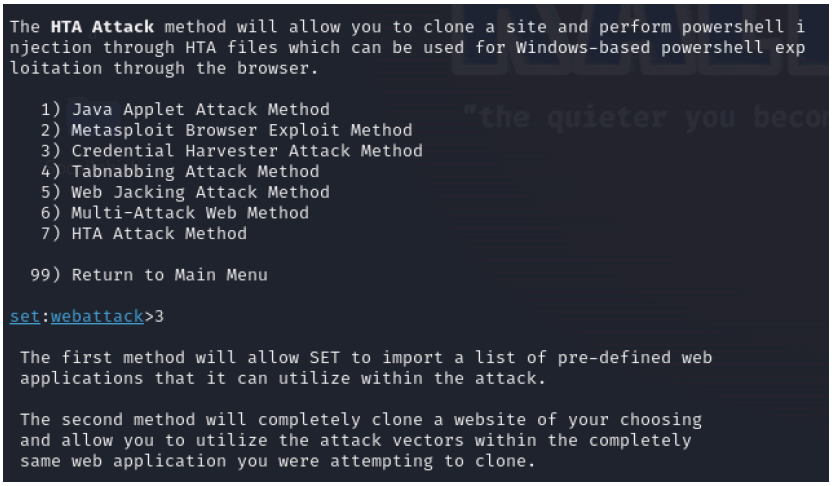


Figure-10

Then I have selected the “web templates” option. This allows me to various selected templates and here I have chosen the Google template. They are demonstrated in figure 11 and figure-12.

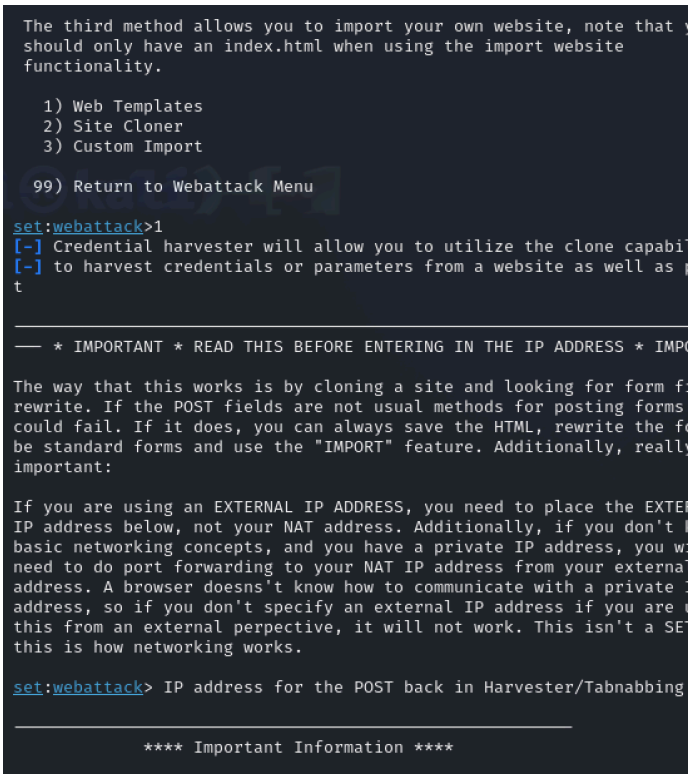


Figure-11



Figure-12

The given IP address directed me to the google website which is fake. However, as showing in figure-13, then I have given the credentials, the username and password are showing in the command prompt.

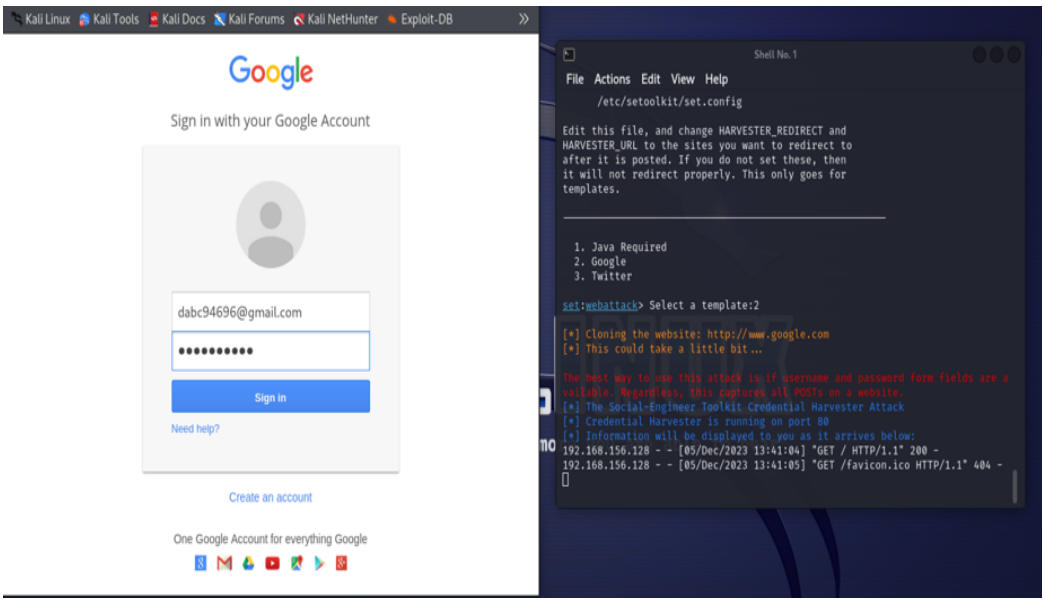


Figure-13

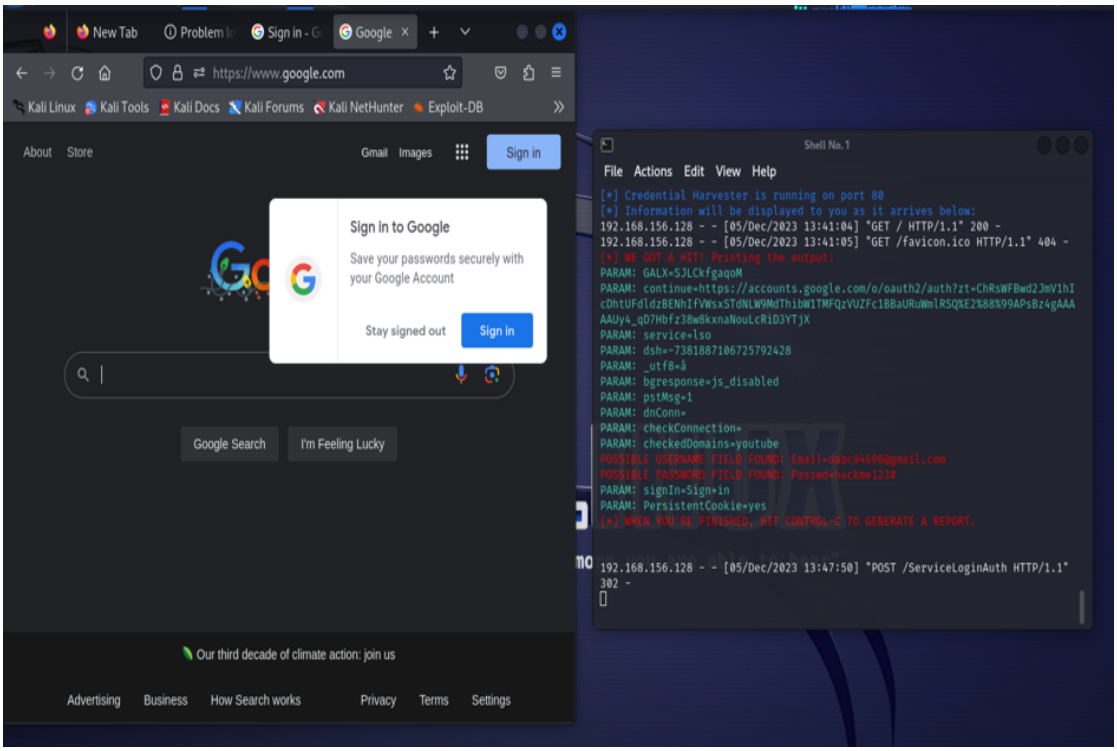


Figure-14

After giving the credentials, it takes back to the real Google website to again sign in (shown in figure-14).

ii. **Zphisher:** Due to its simplicity of use and learnability, beginners may make the most of this phishing simulation tool, which is among the best accessible. It has many features. Such as port supporting tools, less space, docker support and 30+ sites to hack.

Firstly, many dependencies should be installed to run it smoothly. In the beginning, have choose to phish the GitHub website. Then it created 3 URL where I have chosen the third URL which is something like github.com (shown in figure 15,16 &17).



Figure-15

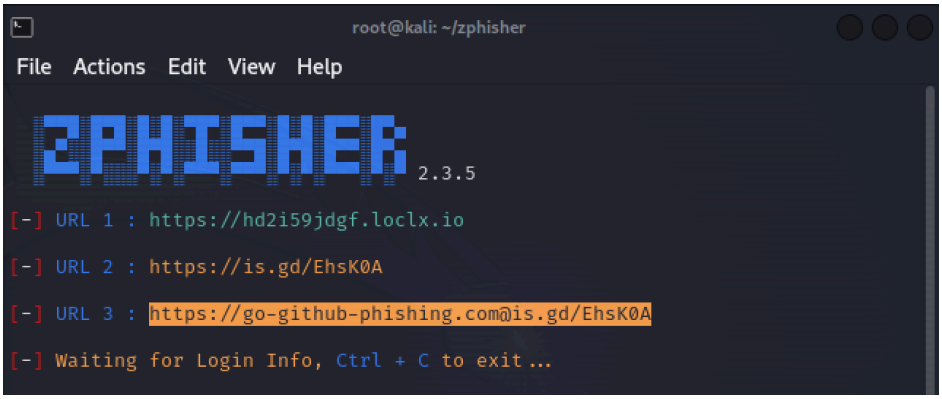


Figure-16

The given link directed me to GitHub phishing site to sign in. So whenever user will give his/her credentials, it will directed back to the original site and records the victim IP, username , password.

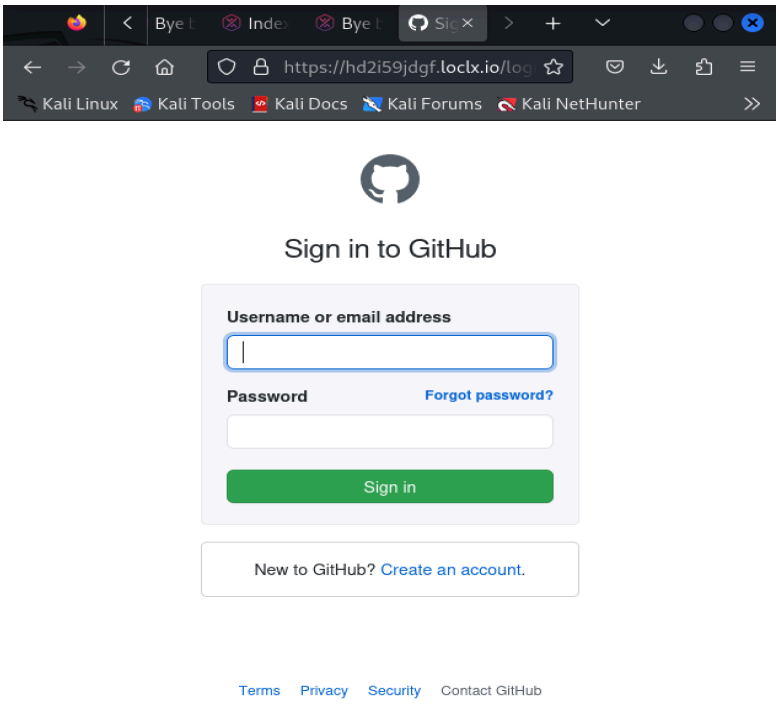


Figure-17

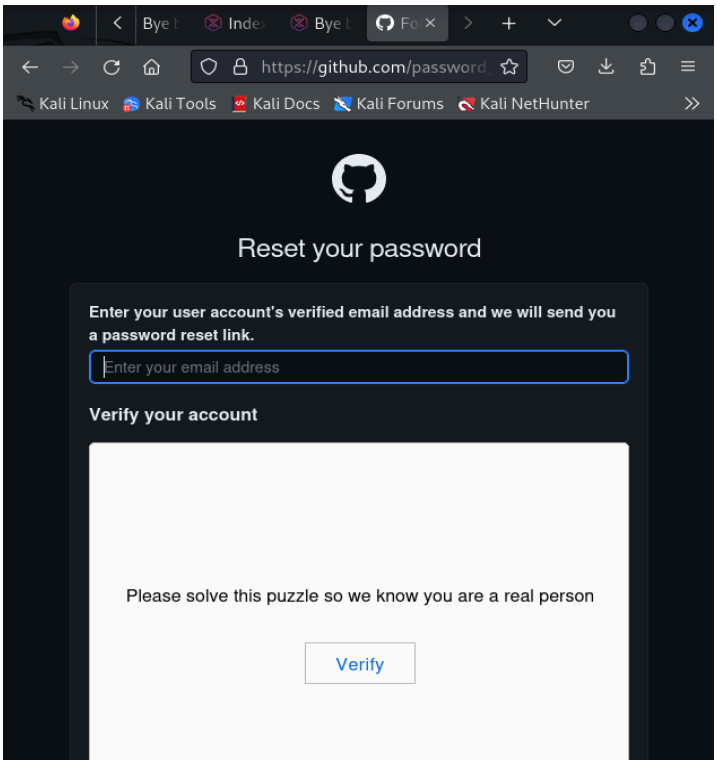


Figure-18

So, after giving the credentials, it directed to original GitHub website and in the Zphisher command panel(figure-18), we can easily see the victim's Ip, name etc(figure-19).



Figure-19

Netflix Phishing Site: Similarly, I have done the Netflix phishing site. Here, the difference is to use the localhost address with the port number. Both worked the same way and redirected to the phishing website.

In the same process, the credentials are stored in the Zphisher database. All are illustrated in figure-20, figure-21 and figure-22.

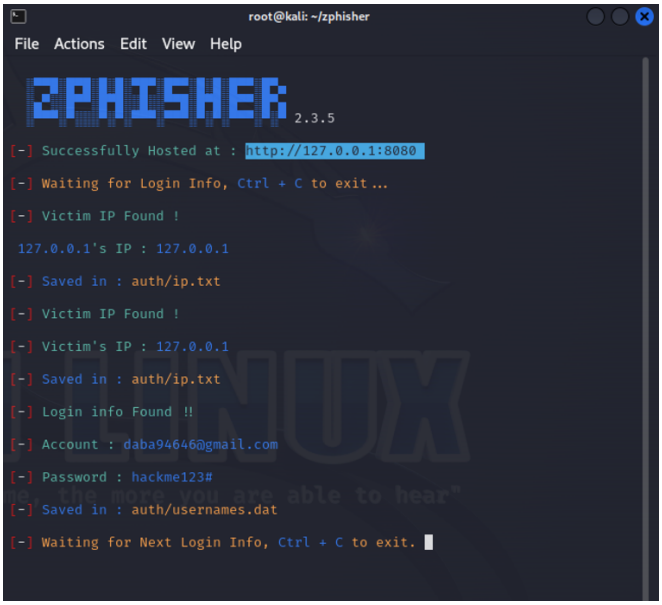


Figure-20

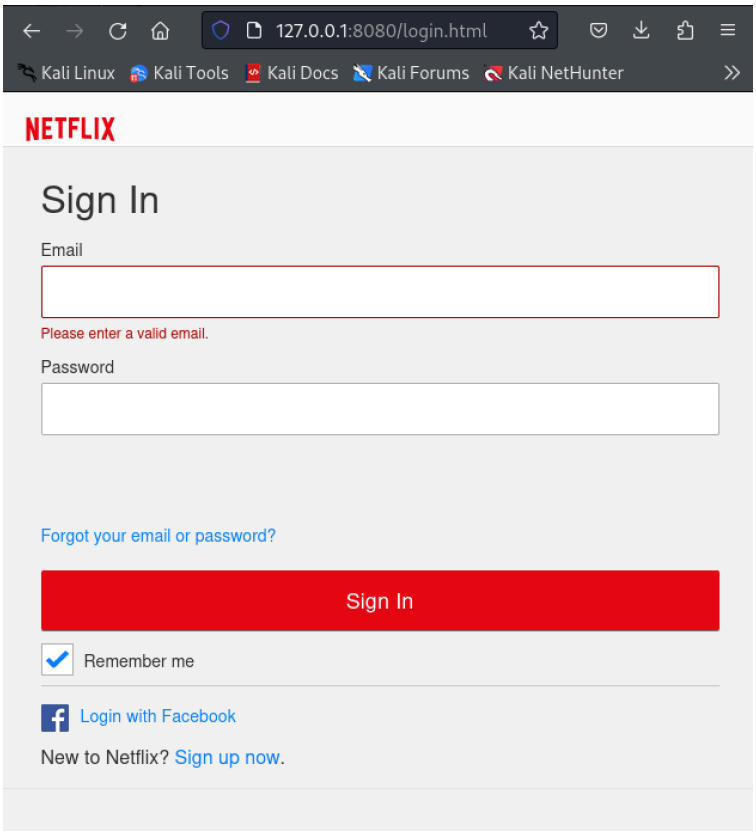


Figure-21

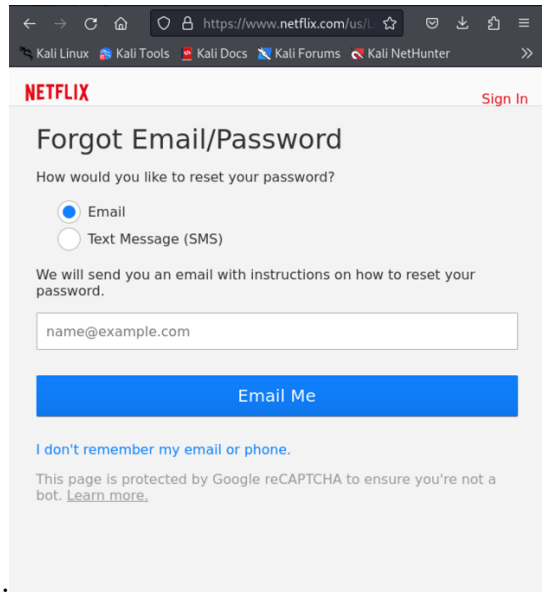


Figure-22

iii. **Socialphish:** Socialphish is one of the classic phishing tools for convincing results with social media phishing, and it's popular for its customized links. It's installation requires Pip3 and Python 3.6 or above. Eventually, the greatest benefit is that with only a few clicks, one can replicate social networking sites and generate a link for a password-harvesting attack.

Spotify Phishing: For phishing this website, I have used SSH tunneling, and default port is 3333. However, then this tool generates a link which is demonstrated in figure-23.

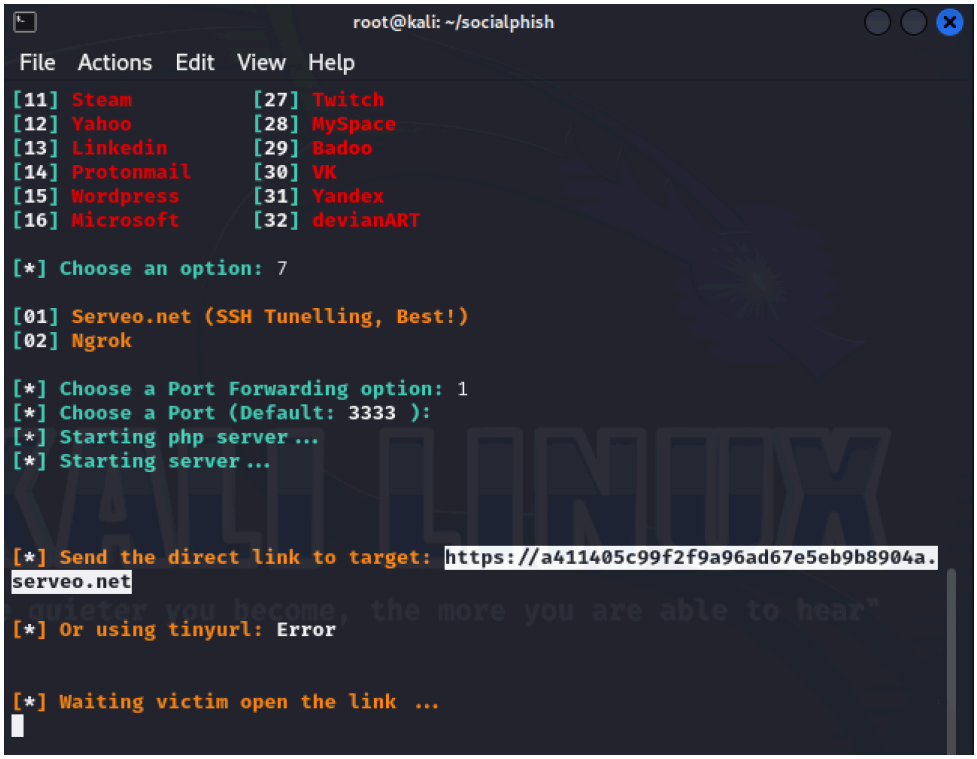


Figure-23

Clicking this link, user will be directed to the Spotify phishing site and there's no way to understand that it is a fake website(figure-24).

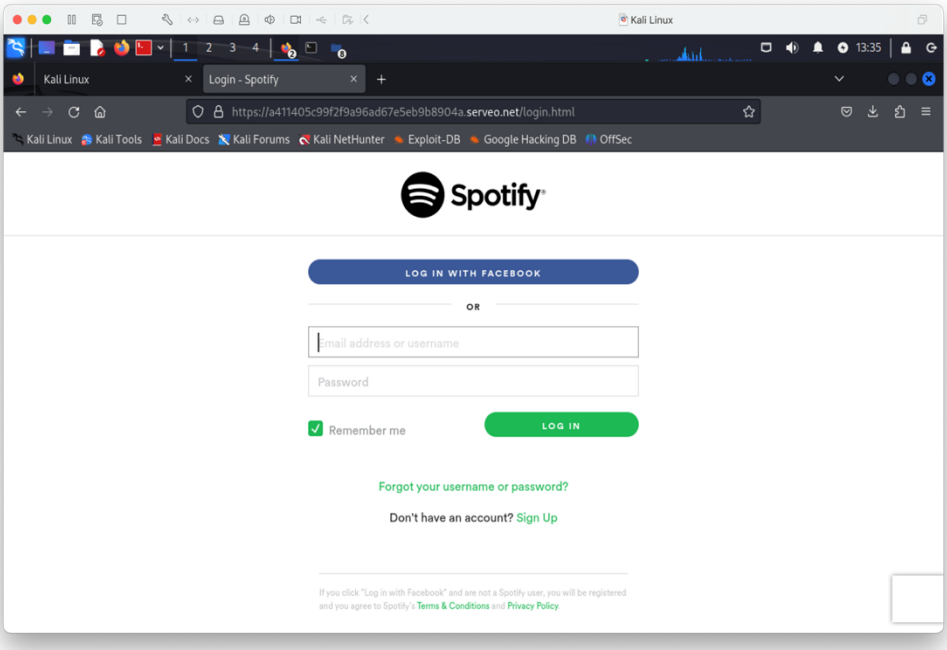


Figure-24

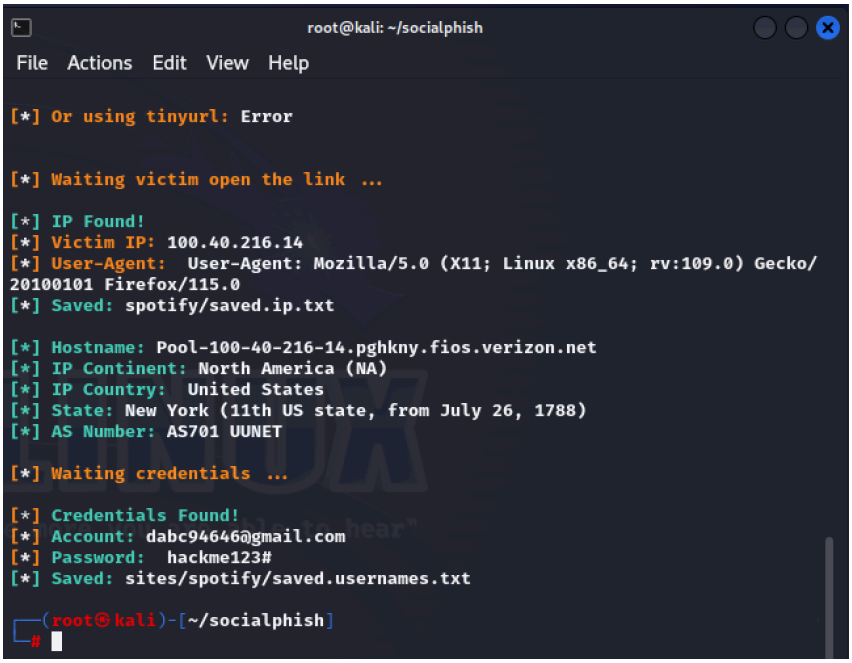


Figure-25

The Victims IP, user agent, country, state, password, account everything is stored in Socialphish website.

Google phishing site: Here, I have done the phishing website of Google.

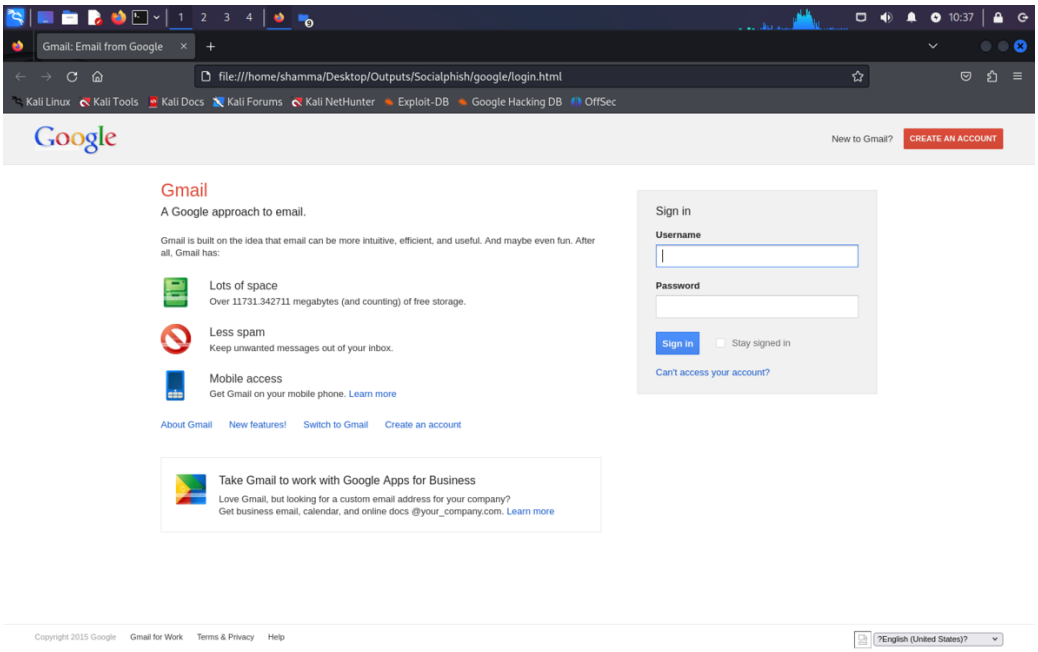


Figure-26

iv. **Evilginx2:** Utilizing phishing and cookie fraud, Evilginx2 is a man-in-the-middle attack tool that allows us to get outside of 2-factor authentication and access users' accounts. Evilginx2 is the updated version of Evilginx.

Using config command, I have configured the phishlets and set the unauthorized URL.

```
: config

 domain          : yee.xyz
 external_ipv4   : 52.80.66.128
 bind_ipv4       :
 https_port      : 443
 dns_port        : 53
 unauth_url      : https://facebook.com
```

Figure-27

However, to run it smoothly, many phishlets should be created and configured. I have only configured the example phishlet which is customized. It needs to be enabled and lure must be created for better working. Lure sets the process is and generates a URL. The given URL takes to phishing website. Below figure-28 is the illustration of this.

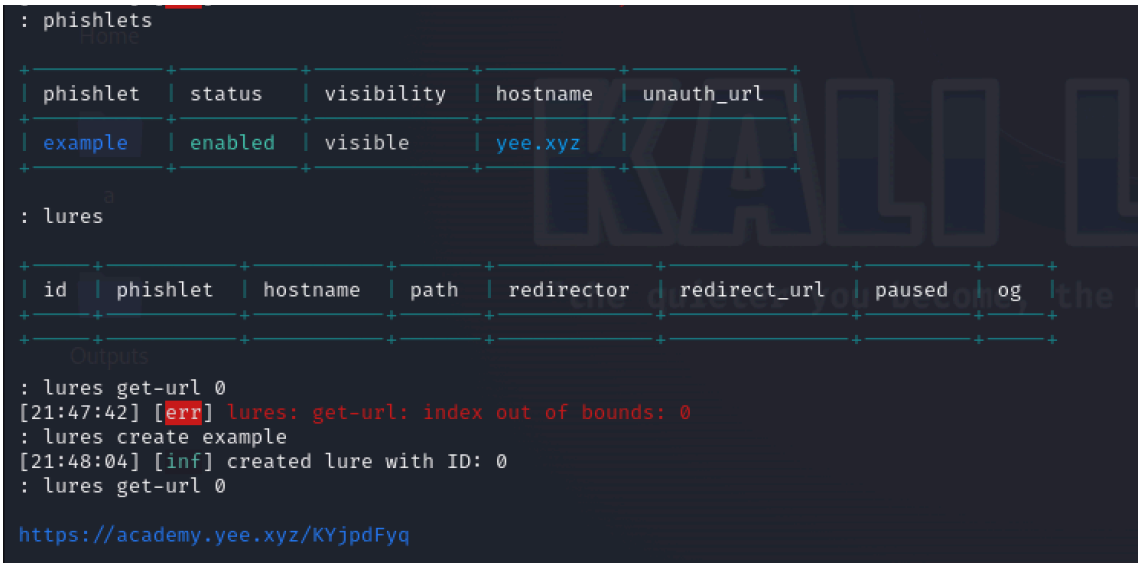


Figure-28

v. **Gophish:** For using penetration testers in businesses, Gophish is an open-source e-fishing toolkit. The information can be retrieved for planned occasions and timelines that monitor email opens, credentials supplied, link clicks, and other activities.

Step 1: Firstly, I have configured the Gophish in the local host. It requires me to create a Gophish account.

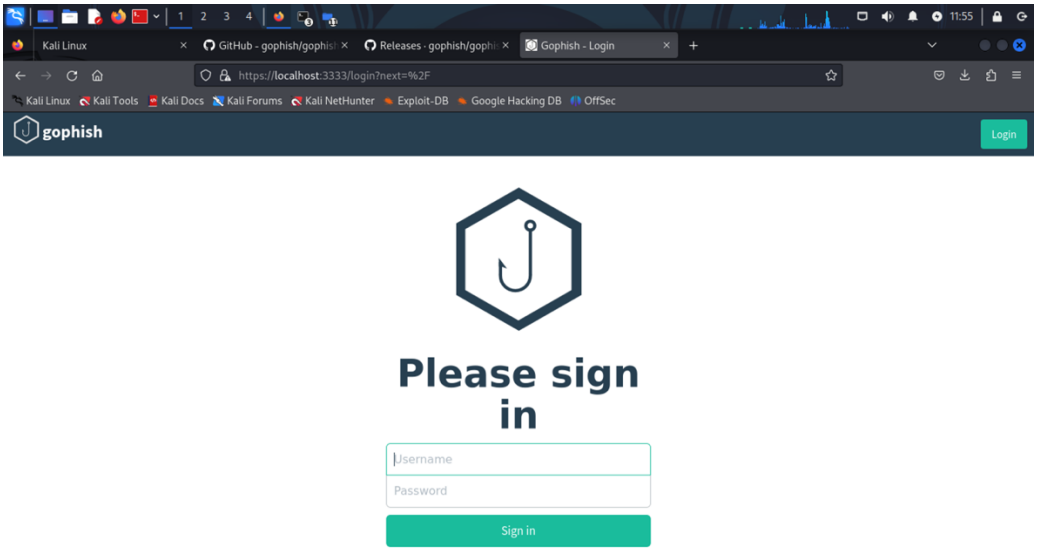


Figure-29

Step 2: Now I have created a sending profile for the testing. I have set the SMTP server as Gmail and port number 587.

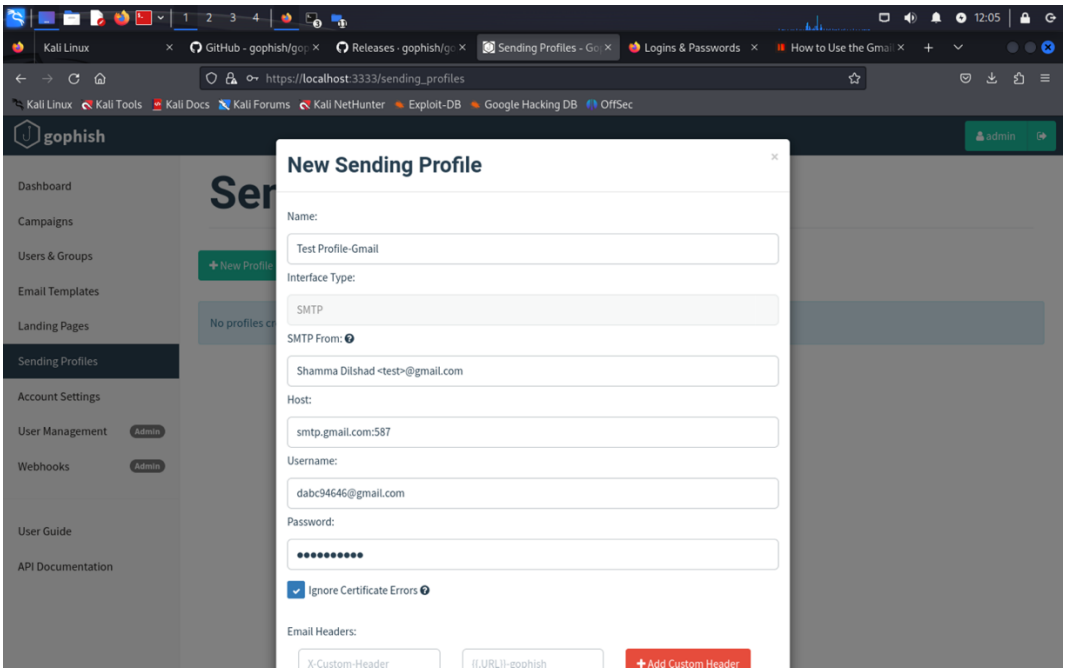


Figure-30

Step 3: Then I have given the email address and send the test email to see if it is really working.

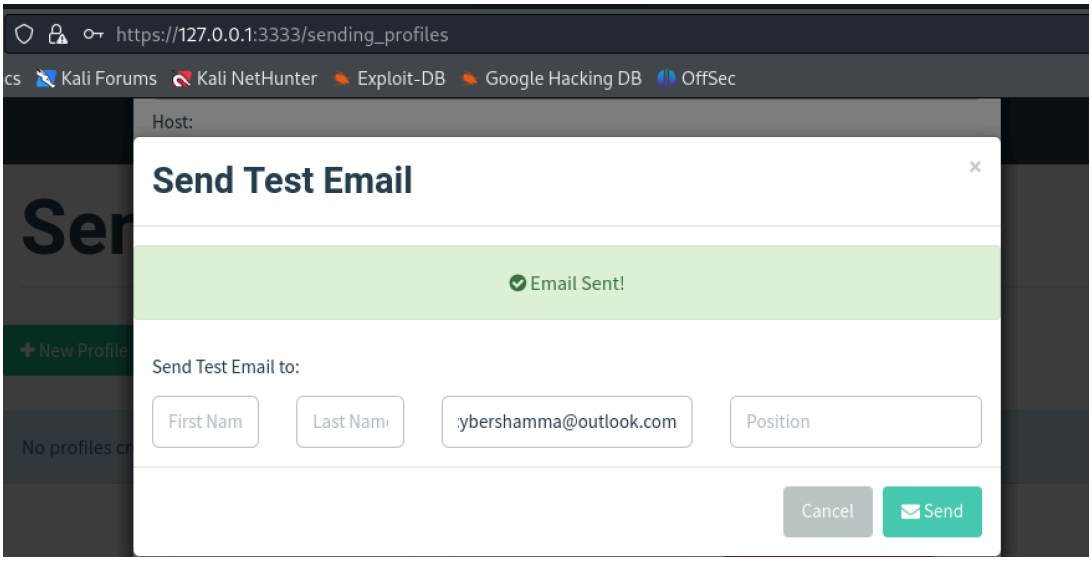


Figure-30

The email was sent successfully (figure-31).

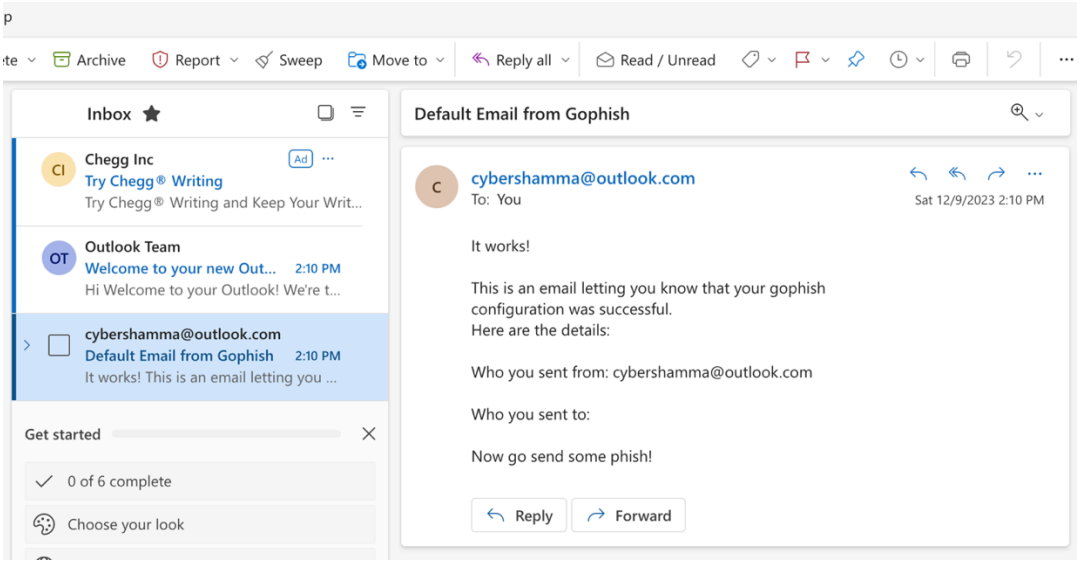


Figure-31

Step 4: After that I have created landing pages and imported a phishing Facebook site which will look quite real.

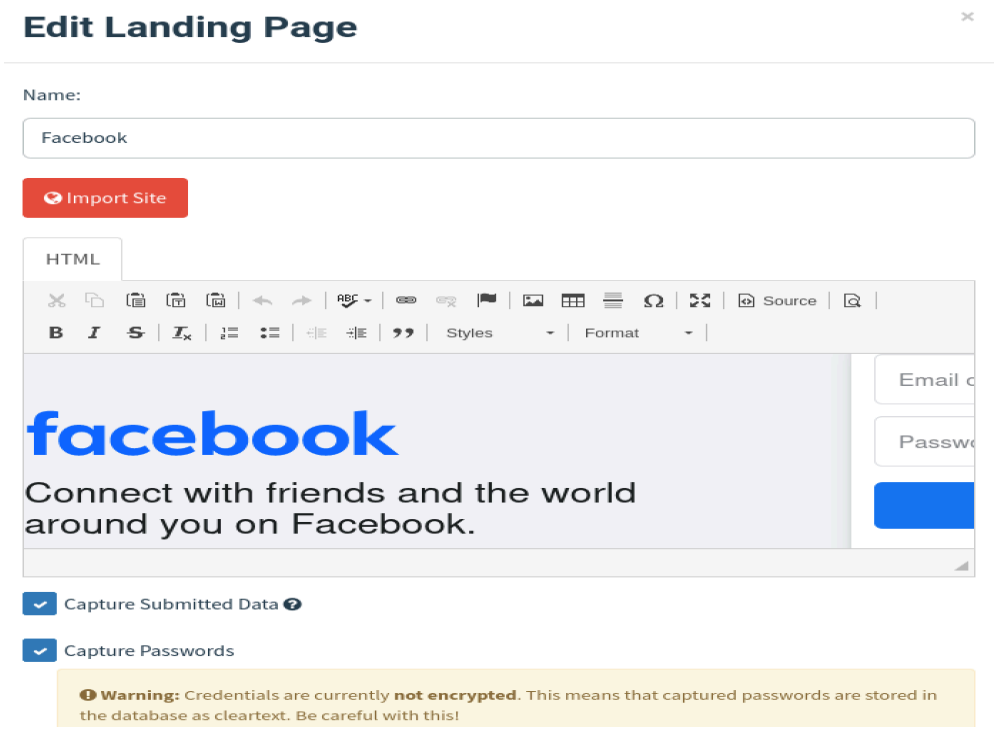


Figure-32

I have also enabled the feature of capturing submitted data and passwords.

Step 5: Then in the template, I have selected a real mail and copy the sources in the text here.

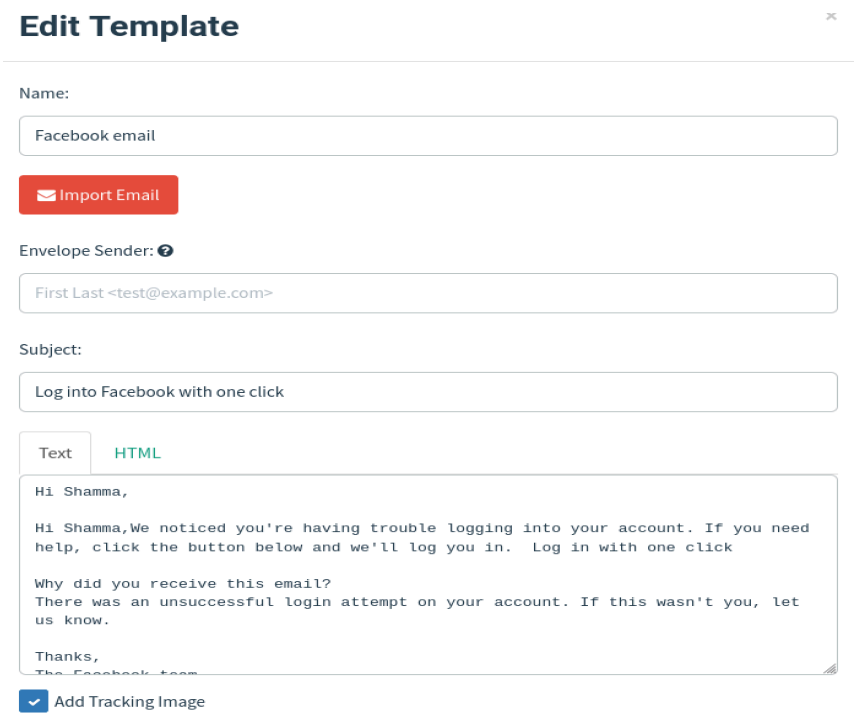


Figure-33

Step 6: Then I have created a group in the below fihure-34.

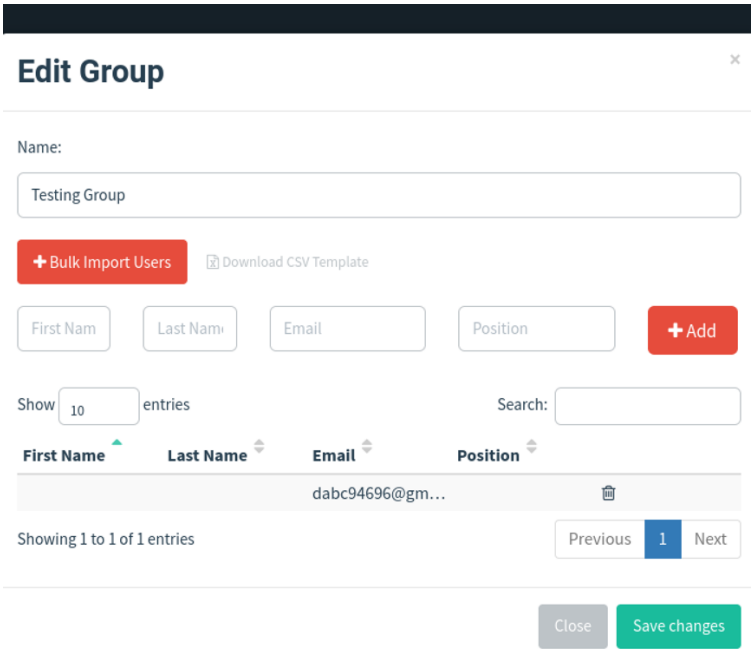


Figure-34

Step 6: In this step, I have launched the campaign to send the phishing email.

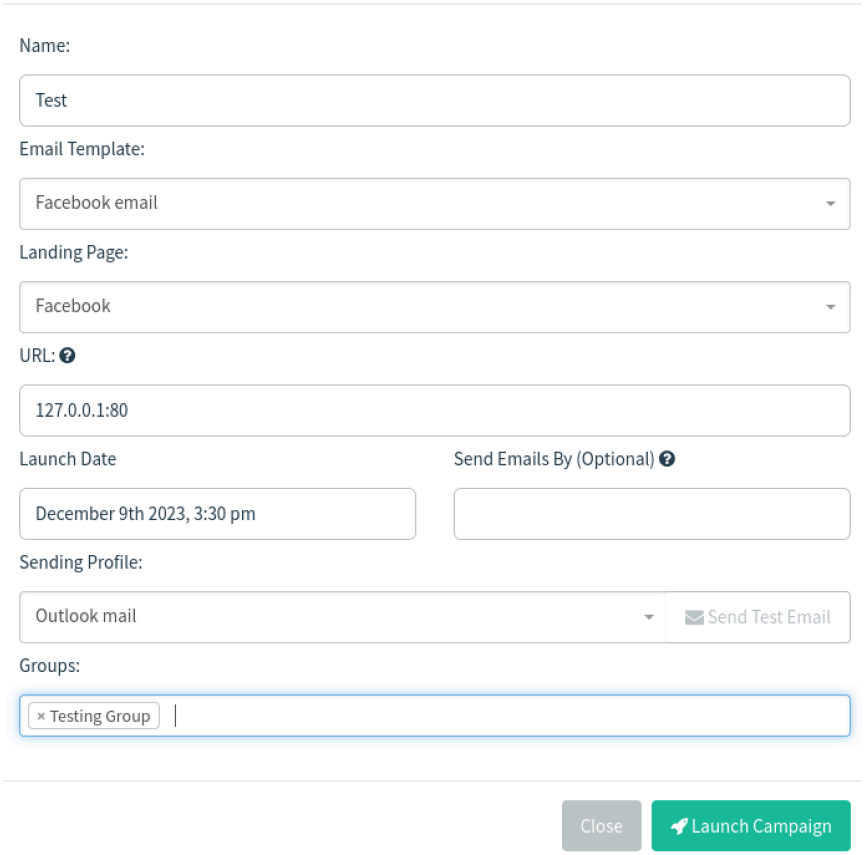


Figure-35

Step 7: Then I have gone to the email inbox to see if it has been received or not.

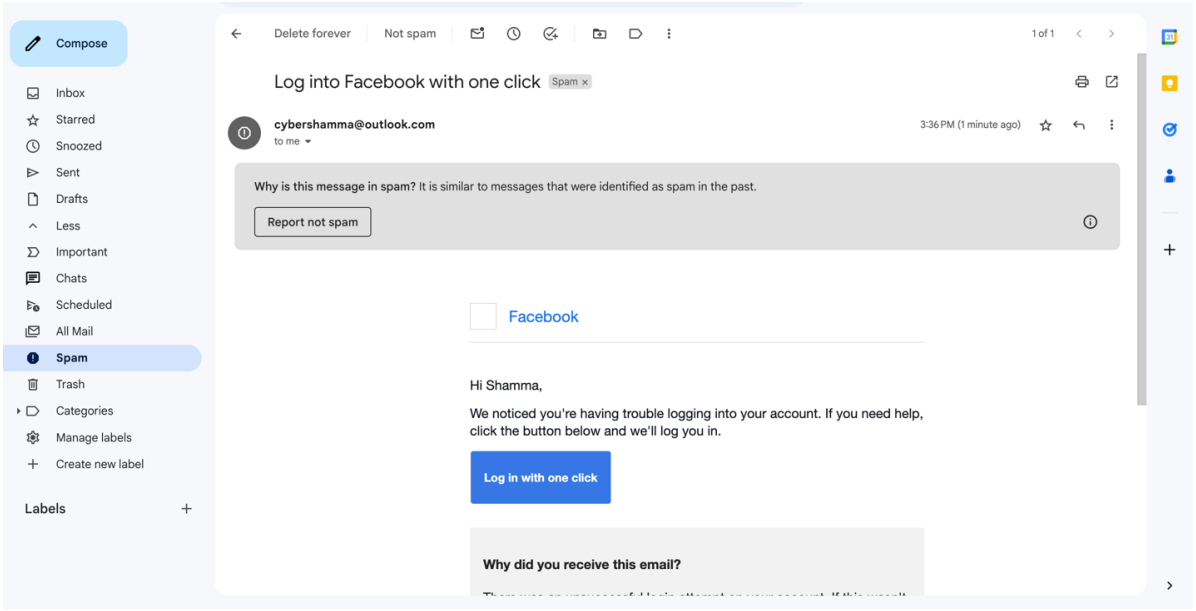


Figure-36

Step 8: Clicking the link, it redirected me to the phishing site of Facebook.

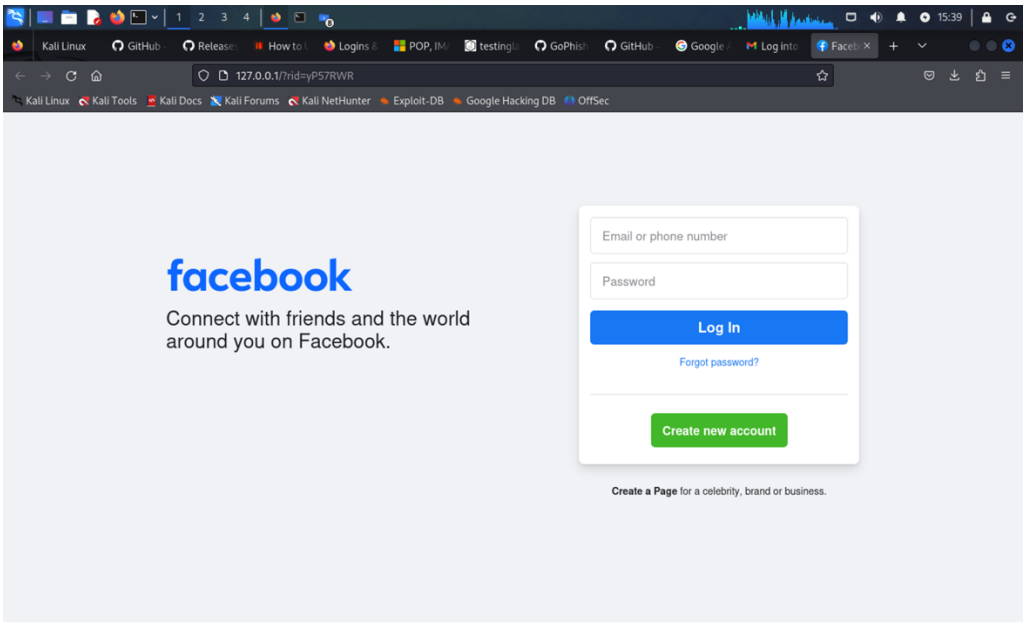


Figure-37

The Dashboard of GoPhish: Here the timeline and submitted data of victim is being gathered and stored.

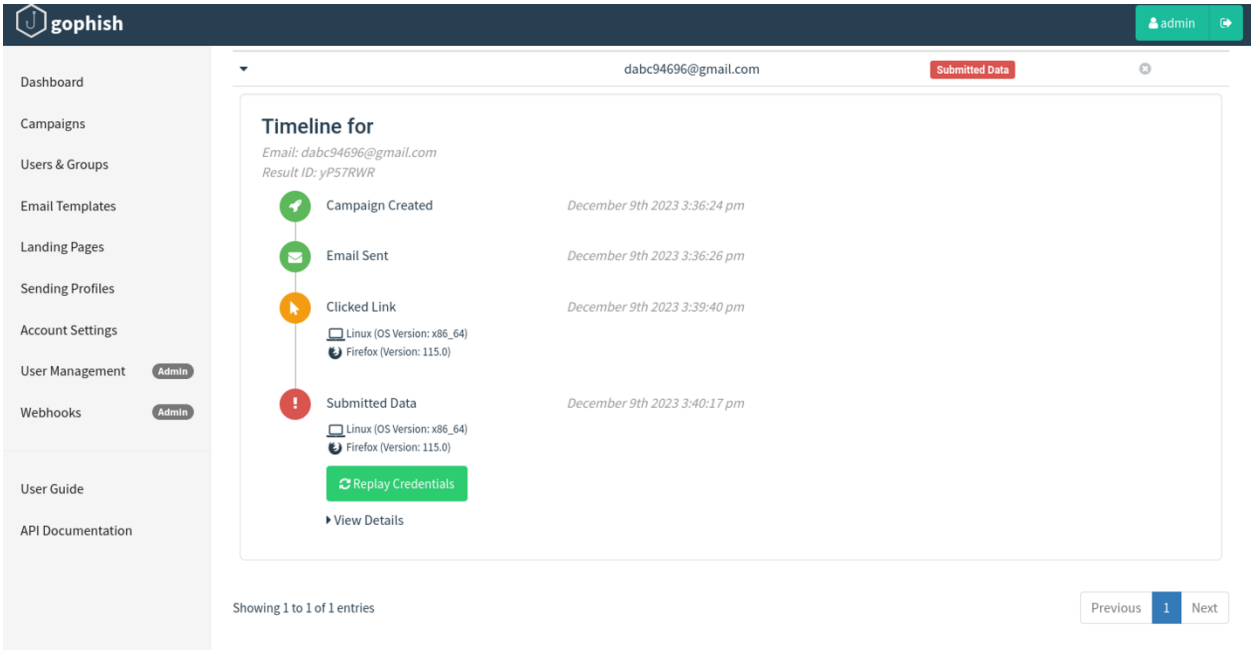


Figure-38



Figure-39

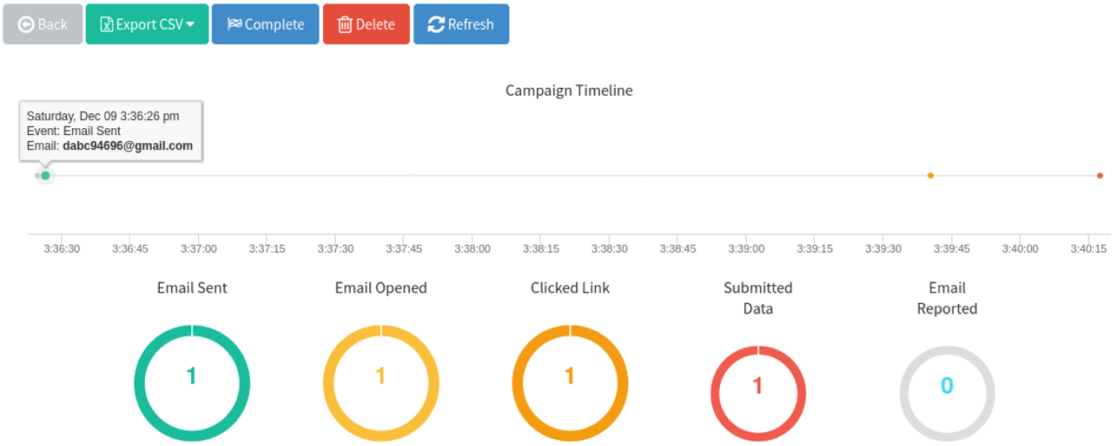


Figure-40

In the above figure-40, total sent email, clicked link, open email information is gathered.

vi. **ShellPhish:** Shellphish is the older version of Zphisher. However, it also works well for Wide Area Network (WAN) data collecting.

By Serveo.net:

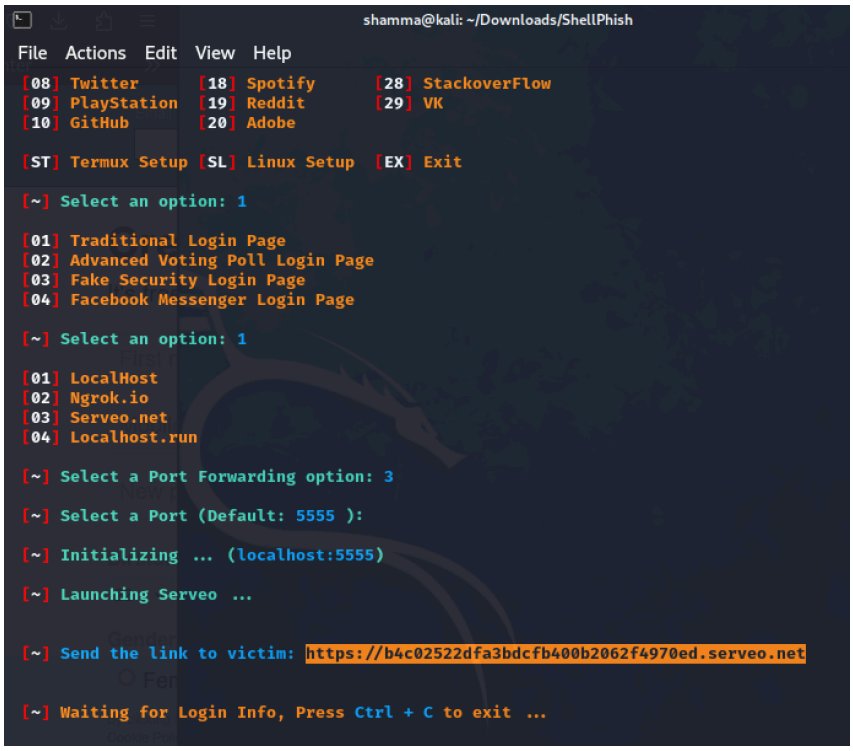


Figure-41

By Localhost:

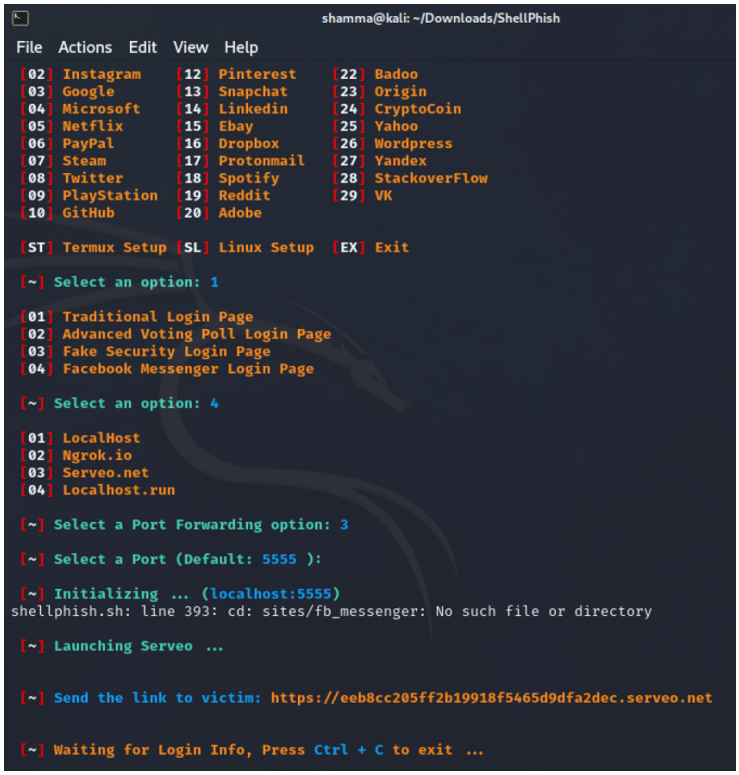


Figure-42

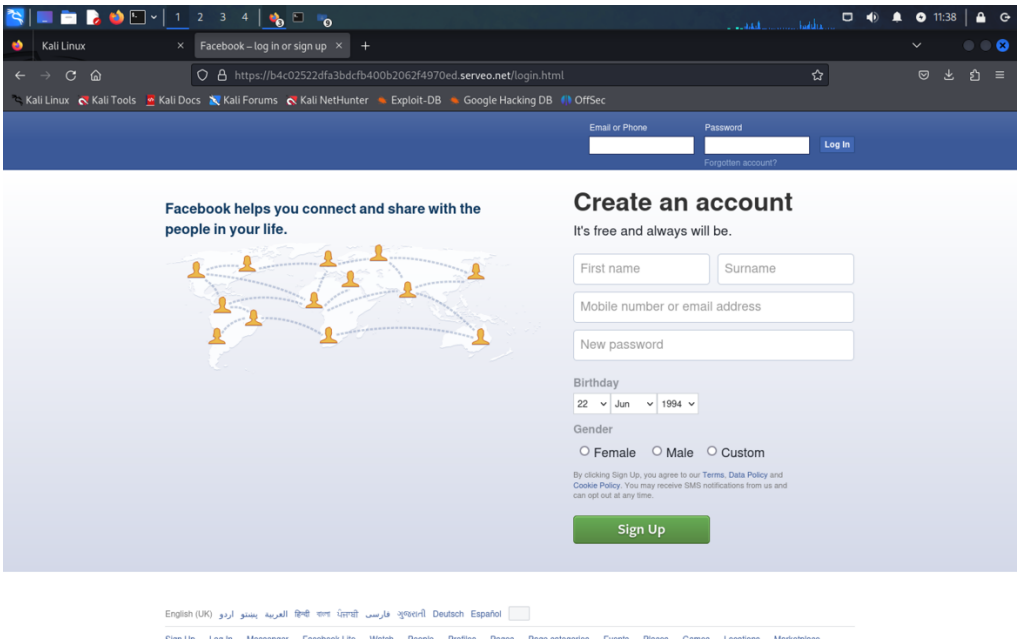


Figure-43

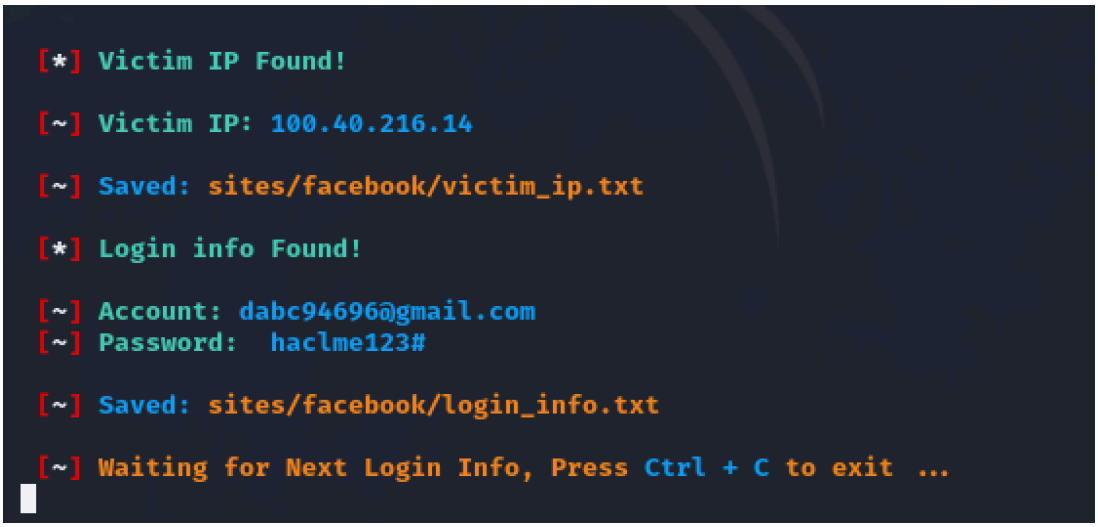


Figure-44

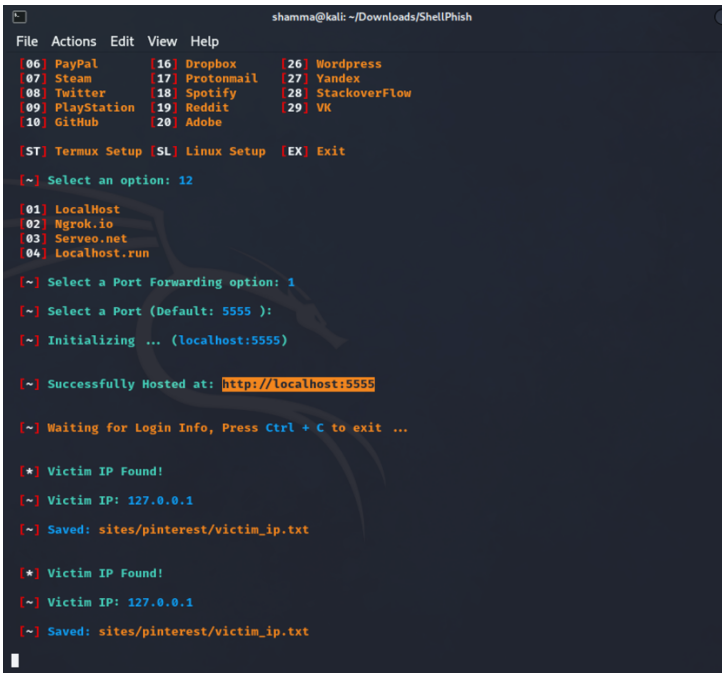


Figure-45

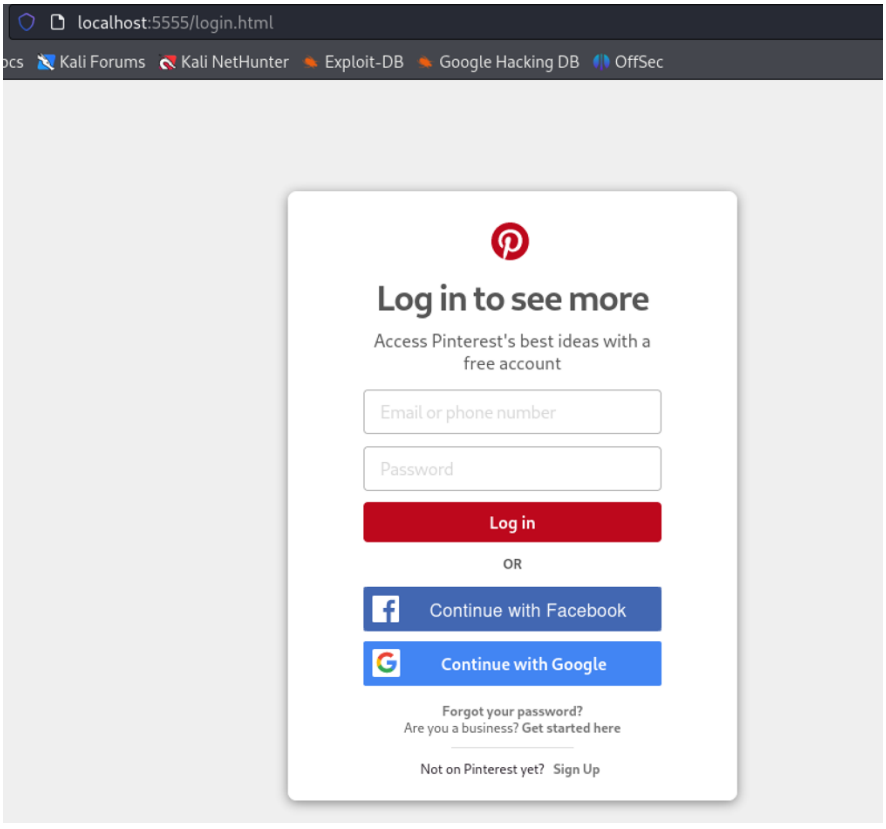


Figure-46

In the above illustration of figure-41 to figure-46, I have phished the website Facebook and Pinterest by localhost and serveo.net. Then it stores the credentials in the database.

Conclusion:

This lab experiment has made clear how important it is to take precautions in order to stop phishing scams. Through the use technological solutions, continuous testing, and user education, businesses may significantly reduce their vulnerability to phishing efforts. The phishing tools simulation have shown us the scenario of how easy it can be trick the victims into phishing and giving their personal information. This cyberthreat is harmful for all the sectors people in society whoever involves with the technology. The knowledge obtained through this inquiry supports the continuous endeavor to strengthen cybersecurity procedures and build a more secure digital environment. Because this is our world, and we need to protect it.

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